

24 }

Listing 8.3: Auth Context Provider

8.4 Firestore Security Rules

```
1 rules_version = '2';
2 service cloud.firestore {
3   match /databases/{database}/documents {
4     function isAuthenticated() {
5       return request.auth != null;
6     }
7
8     match /users/{userId} {
9       allow read: if isAuthenticated();
10      allow create: if isAuthenticated()
11                    && request.auth.uid == userId;
12      allow update: if isAuthenticated();
13      allow delete: if false;
14    }
15
16    match /listings/{listingId} {
17      allow read: if isAuthenticated();
18      allow create: if isAuthenticated();
19      allow update: if isAuthenticated();
20      allow delete: if isAuthenticated()
21                    && request.auth.uid == resource.data.sellerId
22                    ;
23    }
24  }
```

Listing 8.4: Firestore Security Rules (firestore.rules)

8.5 Carbon Credit Calculation

```
1 // Carbon factors (kg CO2 saved per MT of waste diverted)
2 export const CARBON_FACTORS: Record<string, number> = {
3   'Rice Husk': 1320,
4   'Rice Residue': 1280,
5   'Wheat Stubble': 1150,
6   'Sugarcane Bagasse': 980,
7   'Cotton Stalks': 1050,
```